

BONE STORIES - EDUCATIONAL KIT

Activity 2

Associate an astragalus with an animal

A comparative investigation to describe a bone, bring together clues and defend a hypothesis without transforming a proposition into certainty.

AUDIENCE

8-12 years old

DURATION

45-60 mins

CONTENT

Guide, sheet, maps, trails

LENGTH

18 pages

In this kit

1. Facilitator guide
2. Learner sheet
3. Animal cards
4. Neutral locations A-D
5. Method correction
6. Sources and credits

Pilot version - corpus of astragali A-D, identification and final correction to be validated.

Recommended printing: sheet for each student; animal cards and A-D cards for each group; guide and correction for adults. Cutting cards. Front side recommended for cards, double-sided long edge for the rest.

Color: recommended for animals, but the names and frames make the media usable in grayscale.

Version: 1.0 pilot - July 20, 2026.

BONE STORIES - ACTIVITY 2

Facilitator guide

Associate an astragalus with an animal

Guide to learning how to argue a possible association, without requiring expert identification.

TYPE

Facilitator guide

AUDIENCE

Supervising adults

DURATION

45-60 mins

VERSION

Pilot 1.0

Pilot version - corpus of astragali A-D, identification and final correction to be validated.

1. In brief

In groups, the students describe four bones A-D, compare them to possible animals then formulate a reasoned association. In this version, cards A-D are neutral locations: the session trains the method, not the identification of species.

2. Objectives

- Describe a shape without guessing.
- Compare multiple visible clues.
- Formulate a hypothesis with a degree of certainty.
- Name the references needed to verify.

3. Expected success

Successful

Two clues support the association and one uncertainty is named.

In progress

An association is proposed with a single clue.

To resume

The choice rests on the assumed size or chance.

4. Duration and material

Context	Duration	Material
Class	45-60 mins	Sheet, animal cards, A-D cards, pencils.
Museum	30-45 mins	Replicas, educational bones or validated photographs; animal cards.

5. Quick preparation

À PRÉPARER

Cut out the animal cards. Use cards A-D only as templates until the scientific corpus is integrated. Do not prepare a specific response.

À DIRE

“Your job is not to guess. You must defend a hypothesis and say what is missing to verify it. »

6. Detailed procedure

Time	Adult action	Student action	Purpose
5 mins	Present the mission and caution vocabulary.	Read the degrees “possible”, “probable”, “to be verified”.	Lay down the rules of the game.
10 mins	Distribute an A-D card without an animal.	Describe shape, reliefs and surfaces.	Separate observation and attribution.
10 mins	Deal out the animal cards.	Compare locomotion, proportions and organization of the limb.	Build tracks.
15 mins	Request two clues and one limit.	Choose a possible association and argue it.	Explain the reasoning.
10 mins	Put the groups in debate.	Compare the propositions and the missing information.	Understand validation.

7. Questions and reminders

Observe

- What is wide or narrow?
- What surfaces appear articular?
- Which view should be added?

Argue

- Does your clue exclude an animal?
- What other explanation is possible?
- What measurement would help verify?

8. Responses or expected elements

No A-D attribution is expected in this release. The expected productions are verifiable descriptions, an association formulated as a hypothesis, at least two clues and a list of missing information: orientation, scale, species, provenance and comparison to a reference corpus.

9. Differentiation

Need	Adaptation
Help	Reduce to two animals and provide the word bank.
Lighten the writing	Have the argument recorded or dictated.
Go deeper	Have the clues classified from the weakest to the strongest.

10. Museum adaptation

Only use a manipulatable object if museum rules allow it. Otherwise, use an oriented photograph, showcase or replica. Adapt the species to the corpus actually available; never let people believe that a museum necessarily owns the collections mentioned.

11. Quick assessment

Request a sentence comprising four elements: possible association, clue 1, clue 2, information to be verified. Evaluate the quality of the reasoning rather than the name chosen.

12. Common mistakes

À NE PAS AFFIRMER

Do not present the size of the animal as a sufficient criterion. Do not announce cash for empty A-D cards. Do not confuse resemblance and identification.

- Choose your favorite animal.
- Deduce the species from a single curve.
- Forget orientation and scale.
- Use “certain” without reference.

13. Sources and limits

Zeder and Lapham (2010) show the limits of isolated morphological criteria for distinguishing certain postcranial bones of sheep and goats. A reliable attribution requires a reference corpus, an orientation, a scale and zooarchaeological validation. The bat remains an extension, outside the heart of the activity.

14. Current status and remaining validations

À VÉRIFIER

Corpus A-D not integrated. Species, photographs, metadata and final correction must be validated by a person competent in zooarchaeology. The interim scientific media documented in the redesign is explicitly excluded from these PDFs.

Print plan

Each student: student sheet.

Each group: a game of animals and A-D cards.

Adult: guide and method correction.

Cutting: single-sided cards; 120-160 g paper if possible.

BONE STORIES - ACTIVITY 2

Learner sheet

Associate a bone with an animal

TYPE

Learner sheet

AUDIENCE

8-12 years old

DURATION

45-60 mins

VERSION

Pilot 1.0

Pilot version - real bones A-D will be added after scientific validation.

First name:

Date:

Class/group:

Your mission

Compare a bone A-D with several animals, propose a possible association and explain what still needs to be checked.

Objective: learn to argue without guessing and to say when an answer remains uncertain.

Material: an A-D card, the animal cards and a pencil.

1. Observe the bone without naming it

General form	Reliefs and hollows	Smooth surfaces	Question to ask

Word bank: wide - narrow - elongated - rounded - hollow - hump - articular surface - symmetrical.

2. Compare possible animals

Animal	clue which brings	clue which makes you hesitate
Horse		
Sheep		
Deer		
Pork		

3. Formulate your hypothesis

I suggest associating the _____ card with _____.

clue 1:

clue 2:

My level of confidence: ☐ possible ☐ probable ☐ to be verified

4. Say what is missing

To verify my hypothesis, I would need:

5. Final trace

Write a conclusion that contains the words “I observe”, “I suppose” and “I must check”.

À OBSERVER

An observation describes the bone. A hypothesis proposes an association.
Identification requires references and validation.

BONE STORIES - ACTIVITY 2

Supports and cards

Associate a bone with an animal

Animal cards to cut out and neutral locations for the future A-D corpus.

TYPE Cards and media	AUDIENCE Learner groups
USAGE Printing and cutting	VERSION Pilot 1.0

Pilot version - scientific corpus A-D to be integrated after validation.
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Animal cards

Horse

1

Observe the silhouette and the limb without deducing only the bone.

evelynbelgium, CC BY-SA 2.0, via Wikimedia Commons.

Sheep

2

Small ruminant used as a possibility of comparison.

Fernando Losada Rodriguez, CC BY-SA 4.0, via Wikimedia Commons.

Deer

3

Deer presented without generalization to all species.

Luc Viatour, CC BY-SA 3.0, via Wikimedia Commons.

Pork

4

Whole animal: basis for comparison, not proof of identification.

Joshua Lutz, public domain, via Wikimedia Commons.

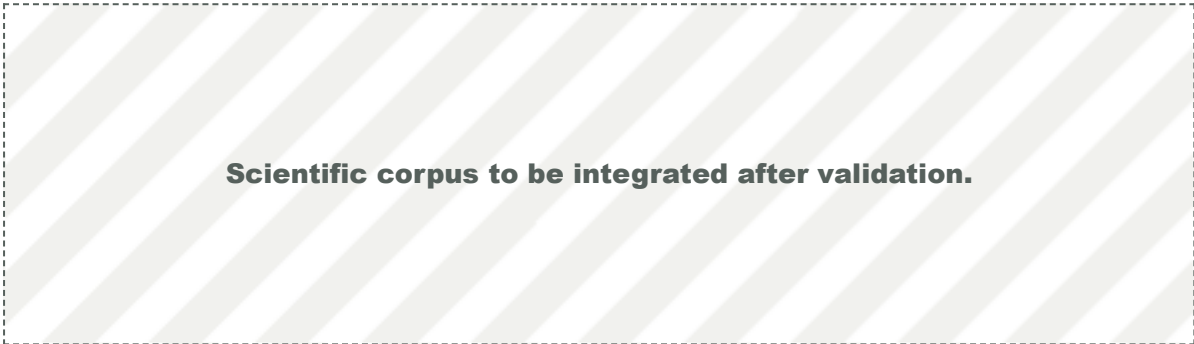
Bone cards A-D - neutral templates

À VÉRIFIER

The photographs are not yet validated. Do not write species names on these cards before integration of the final corpus.

Bone A

A



orientation:

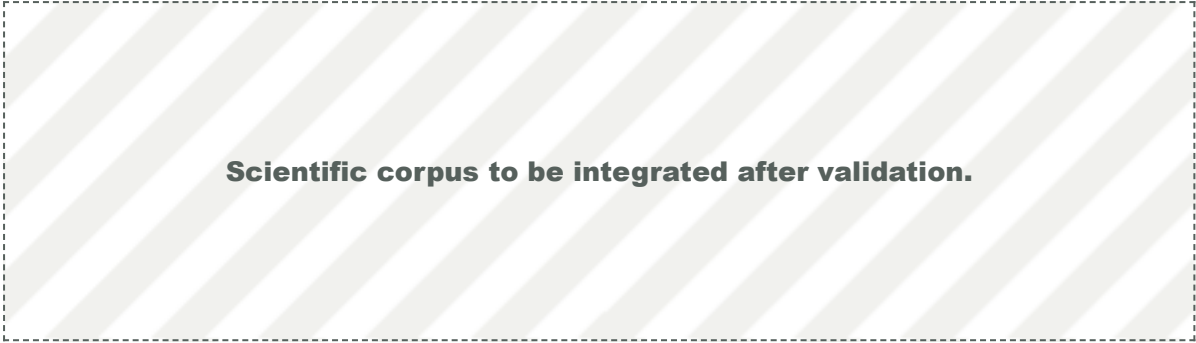
Scale:

Source:

credit:

Bone B

B



orientation: _____

Scale: _____

Source: _____

credit: _____

Bone C

C



orientation: _____

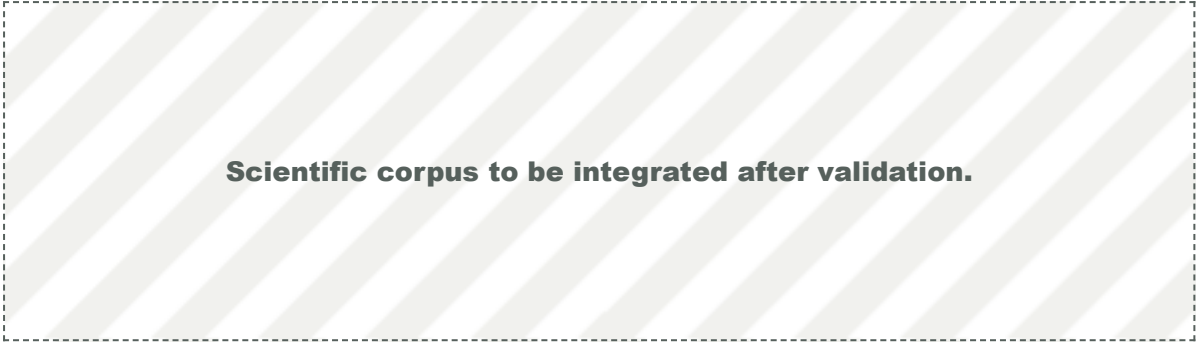
Scale: _____

Source: _____

credit: _____

Bone D

D



orientation: _____

Scale: _____

Source: _____

credit: _____

BONE STORIES - ACTIVITY 2

Method correction

Associate a bone with an animal

TYPE Adult guidance	AUDIENCE Teachers and museum educators
USAGE Group review	VERSION Pilot 1.0

Pilot version - no definitive A-D attribution before validation of the corpus.

Quick correction grid

Step	Expected response	Possible answer	Not to be validated alone
Observation	Several visible characteristics.	Width, reliefs, surfaces, proportions.	Animal name without description.
Comparison	clues linked to the cards.	Organization of the limb, locomotion, proportions.	General size of the animal.
Hypothesis	Association announced as possible.	“Probable according to these two clues”.	“It has to be...”.
Verification	At least one data missing.	orientation, scale, reference corpus.	New unsourced assumption.

Useful phrases

Expected response “The association is possible because I observe two clues, but the species must be verified with a reference corpus. »	Possible answer Two groups can propose different animals if their clues are explained and their limits recognized.
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Common errors

- Confusing whole animal and shape of the bone.
- Forget scale or orientation.
- Taking a resemblance for identification.
- Transform the method answer key into a species key.

Scientific limits

Bones A-D are not present in this version. So there is no correct association to reveal. This answer only evaluates the approach: observe, compare, argue, verify.

À VÉRIFIER

Complete only after archaeozoological validation of the corpus, the notes on each bone and the criteria that can really be used with students.

BONE STORIES - ACTIVITY 2

Sources, credits and limits

Keep with the complete kit.

Animal Pictures

- Horse: evelynbelgium, CC BY-SA 2.0, via Wikimedia Commons.
- Sheep: Fernando Losada Rodríguez, CC BY-SA 4.0, via Wikimedia Commons.
- Deer: Luc Viatour, CC BY-SA 3.0, via Wikimedia Commons.
- Pig: Joshua Lutz, public domain, via Wikimedia Commons.

Scientific references

- Melinda A. Zeder and Heather A. Lapham, "Assessing the reliability of criteria used to identify postcranial bones in sheep and goats," *Journal of Archaeological Science*, 2010, DOI 10.1016/j.jas.2010.06.032.

Media excluded

The provisional comparative figure of Vuillien et al. is not integrated: its conditions of reuse have not been established. Cards A-D remain neutral until scientific and legal validation.

Status

Method support usable for a supervised test. Corpus, identification, final correction and physical printing test remain to be carried out.

Working translation: specialist scientific, educational and legal wording requires human review.

Traceability : **Activity page, updates and online credits.**